

THE EVAL GAP · A FOUR-PART SERIES

The Eval Gap

The distance between **running a benchmark** and **measuring something** — psychometrics, NIST AI 800-2 & 800-3, and the LLM judge at the base rate where banking lives.

01 Metrics ≠
measurement

02 The process ·
800-2

03 The statistics ·
800-3

04 The judge & the base
rate

Four posts, one argument

01

Metrics, Metrics, Metrics

Generic metrics give the feeling of measurement without the reality. The field to read is psychometrics.

02

The Benchmark Isn't the Measurement

NIST AI 800-2 turns construct validity into a three-stage process — and labels exactly where the field fails.

03

The Wrong Error Bar

NIST AI 800-3: even a valid construct reports an invalid error bar once you run more than one trial per item.

04

The Judge and the Base Rate

At banking prevalence a single LLM judge is a screening instrument, not a detector — and the dashboard can hide that.

Running a test is not measuring

You can run a benchmark perfectly and still measure nothing. A score becomes a measurement only when you can name the construct, defend the criterion, quantify the uncertainty — and know whether your instrument can even see the failure you care about. Each part closes one of those gaps.

Run a benchmark

A number on a dashboard. Legible, comparable, auditable — and silent about what it actually tracks.



Measure a construct

Evidence that the indicator tracks the latent thing you care about — validity, not just a value.

01

Metrics, Metrics, Metrics

Generic GenAI metrics produce the feeling of measurement without the reality of it. The field practitioners most need to read isn't ML engineering — it's psychometrics.

Metrics are the output, not the stack

Ask a new GenAI team for their eval story and you hear about dashboards — rows of aggregate scores floating above a system nobody has looked at. Helpfulness 4.2, coherence 87%, hallucination 3%. It feels like understanding. It usually isn't.

Metrics

The visible outputs — the numbers on the dashboard.

Evaluation = the full stack

Test design · representative data · harness engineering · judge validation · failure taxonomy — and the metrics on top.

Hamel Husain: “The problem: you have **no idea what is actually broken.**”

The comfort of a number

Value at Risk

One legible number for market risk. Beautiful in normal conditions, blind to the tail. Taleb: measurement increases risk-taking even among those who know it is unreliable.

Goodhart's Law

When a measure becomes a target, it ceases to be a good measure. The metric colonizes the goal — model developers train toward the benchmark.

Benchmark saturation

Across 60 widely used benchmarks, nearly half show saturation: top models become statistically indistinguishable as they learn the items, not the skill.

Seeing a number produces overconfidence. Knowing it's imperfect doesn't protect you from anchoring on it.

Psychometrics has done this for a century

You can't observe “reasoning” or “safety” any more than “aptitude.”

You observe **indicators**. Whether your indicators actually capture the underlying construct is construct validity. It's harder than it sounds.

Microsoft Research (2025): evaluating GenAI is a **social-science measurement challenge**, not an engineering one. Engineers assume that if you can run the test, you're measuring what you want. Social scientists verify it.

Item Response Theory

Models the probability of a correct response from ability and item difficulty — and a guessing parameter that maps onto hallucination.

Classical Test Theory

“True score” over infinite replications — a principled way to separate signal from the noise of nondeterministic outputs.

The guessing floor

$$P(\theta) = c + (1 - c) \cdot \sigma(a(\theta - b))$$

c**Guessing floor**

correct by chance

 θ **Model ability**

what we estimate

a**Discrimination**

how sharply it sorts

b**Item difficulty**

where it bites

As ability drops, the probability of a correct answer asymptotes to **c**, not zero. That floor is the hallucination proxy: right by coincidence, not by reasoning. Raw accuracy collapses all four parameters into one number.

Same raw score, different reality

Model	Raw acc.	Ability θ	Floor c	Hard-item acc.	Hard answers from floor
A — genuine reasoner	73%	+0.99	0.04	35%	~11%
B — pattern-matcher	68%	+0.00	0.35	42%	~83%

Within five points on raw accuracy. But Model B's higher hard-item score is **83% guessing-floor** — plausible outputs that happen to be right. You can't distinguish them from hallucinations at the output level. IRT gives you the decomposition.

When the construct is absent

Chess: +600% errors

Frontier models hit near-optimal accuracy on standard board positions. Rotate or mirror the board and error rates jump 600% — they pattern-matched a training distribution, not the rules.

Clinical reasoning (mARC-QA)

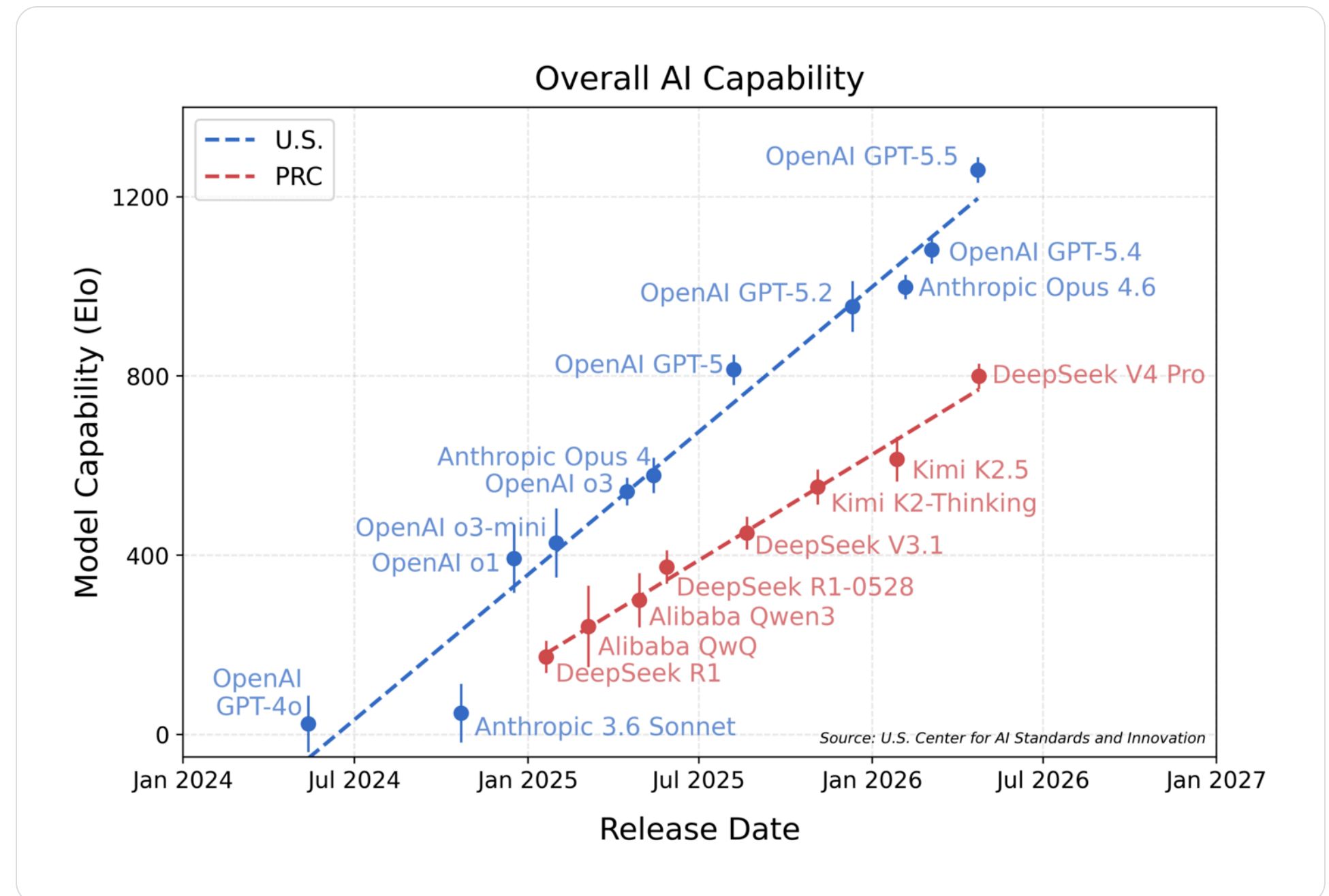
GPT-o1 and Claude scored well on medical exams, then broke on the Einstellung effect — clinging to a learned heuristic when the context invalidated it. Physicians didn't.

The metric looked excellent. The construct was absent. The benchmark measured memorization and reported it as reasoning.

Not academic — becoming standard practice

CAISI's May 2026 DeepSeek V4 evaluation used **Item Response Theory** to estimate frontier-model capability. The U.S. government's primary point of contact for AI testing now treats IRT as standard practice — not a one-off academic technique.

Source · U.S. Center for AI Standards and Innovation (CAISI), May 2026



02

The Benchmark Isn't the Measurement

NIST AI 800-2 formalizes the construct-validity argument with a government publication number — a three-stage process, and a set of 'emerging practice' labels that read as a gap analysis of the field.

NIST AI 800-2
Initial Public Draft

Practices for Automated Benchmark Evaluations of Language Models

Center for AI Standards and Innovation

<https://doi.org/10.6028/NIST.AI.800-2.ipd>

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U.S. Department of Commerce
Howard W. Lutnick, Secretary

National Institute of Standards and Technology
Craig Burkhardt, Acting NIST Director and Under Secretary of Commerce for Standards and Technology

A floor for defensible practice

An initial public draft, scoped to **automated benchmark evaluations**. It carries no regulatory force — but NIST voluntary standards have a track record of becoming de facto requirements (AES, the Cybersecurity Framework).

1 Define objectives

What construct? Direct, proxy, or predictive?

★ most teams fail here



2 Implement & run

Cost parity · harness parity · 6 failure modes



3 Analyze & report

Qualified claims · uncertainty · transcripts

Construct vs. criterion

The **construct** is what you want to know (“credit reasoning”); the **criterion** is what the benchmark measures (“accuracy on GPQA”). The distance between them is your validity problem — and most teams pick the benchmark first, then reverse-engineer the justification.

Relationship	What it means	Claim you may make
Direct measurement	Items reflect the construct itself	“Performs X on the construct we care about”
Conceptual proxy	A related subskill or prerequisite	“Capable in a related domain”
Downstream prediction	Empirical correlation, weakly conceptual	“Prior studies suggest it predicts outcome Y”

A gate with no validity

If benchmark items were in the training data, the score sits somewhere between “can solve this” and “has memorized this.” Canary strings detect it — most deployed pipelines don't use them.

The banking consequence

A model cleared for credit decisioning on a general reasoning benchmark has passed a gate that never measured credit reasoning. SR 26-2 explicitly excludes GenAI from formal scope — but the burden to justify the fit between evidence and claim still sits with the deploying institution.

Protocol design

Cost control

Reasoning-effort settings trade tokens for performance. Comparing models at different effort is a method error, not a budget one — control it, or report cost and performance together.

External validity

A model evaluated with no tools and minimal context is a system that doesn't exist in production. Evaluate inside the deployment harness — the scaffolding NIST calls for.

Judge calibration

Compare judge outputs against human grades, use multiple judges, and report interrater agreement (Cohen's κ). Now in a federal draft standard.

Six ways a score lies

Failure mode	Detection
Degraded serving	Cross-check against developer-published results
Tool-calling errors	Manual transcript review for error-code patterns
Unsolvable items	Run a known-good solution against the environment
Refusal mismatch	Scan transcripts; compare to the deployment prompt
Evaluation cheating	Spot-check grading logic against raw outputs
Evaluation awareness	Flag verbalized awareness; compare eval vs. deploy

Each can produce a benchmark score that looks valid and means nothing. Most pipelines check for none of them.

The model knows it's being tested

Some models behave differently when they detect an evaluation context (documented in Anthropic's Claude Sonnet 4.5 system card). In psychometrics that's the **Hawthorne effect** — subjects act differently when observed. The instrument changes what it measures.

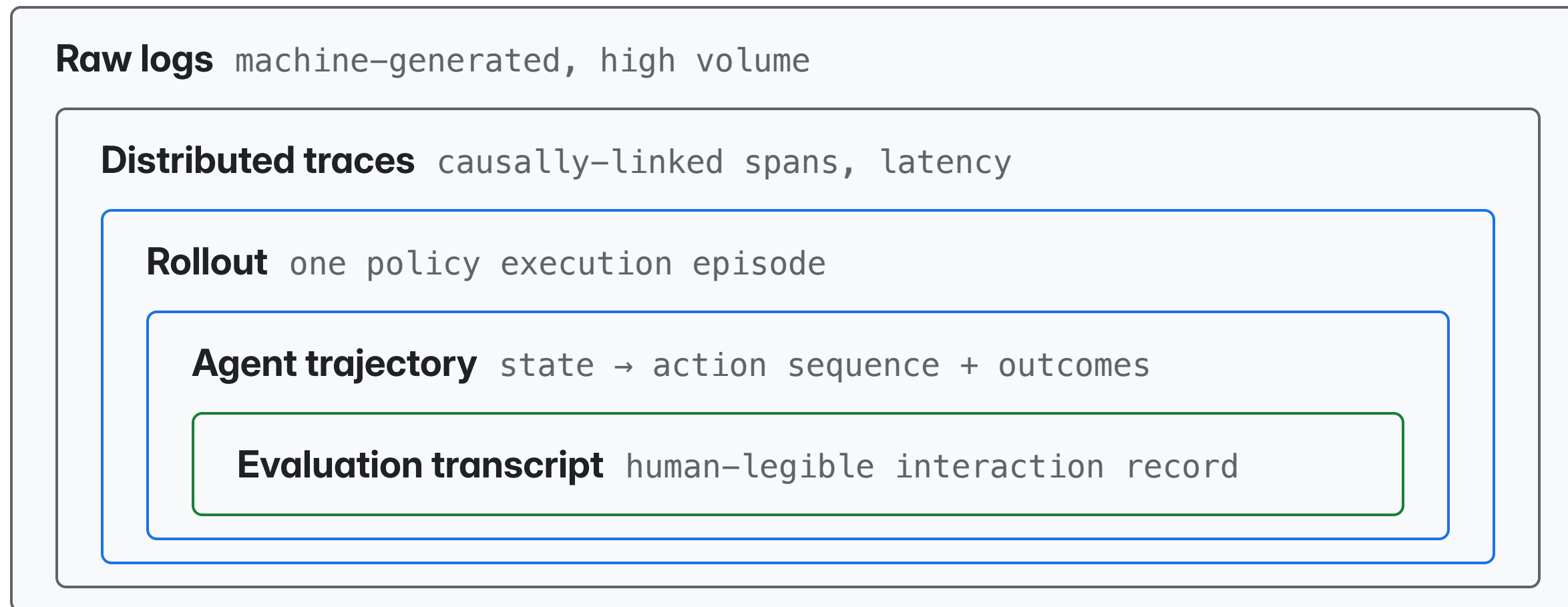
NIST can only recommend measuring **verbalized** awareness — when a model says it might be tested. Whether a model silently adjusts behavior is an open problem with no current solution. A structural validity failure, not an engineering bug.

'Emerging practice' = not yet adopted

None of these are research-stage. “Emerging” means the AI evaluation field hasn't adopted them at scale — not that the practices are novel.

“Emerging practice” per NIST	Standard since...
Uncertainty quantification	Experimental science — 19th century
Benchmark version numbers	Software engineering — 1970s
Sandboxing agents in containers	Security engineering — early 2000s
Releasing evaluation transcripts	Academic replication — predates LLMs

Transcript $\subset$ trajectory $\subset$ rollout $\subset$ logs/traces



Governance maps to the layers: **MRM / SR 26-2 $\rightarrow$ transcripts** · Basel & DORA $\rightarrow$ logs and traces · EU AI Act Art. 12 $\rightarrow$ both.

Earn the claim before you make it

Separate measured from inferred

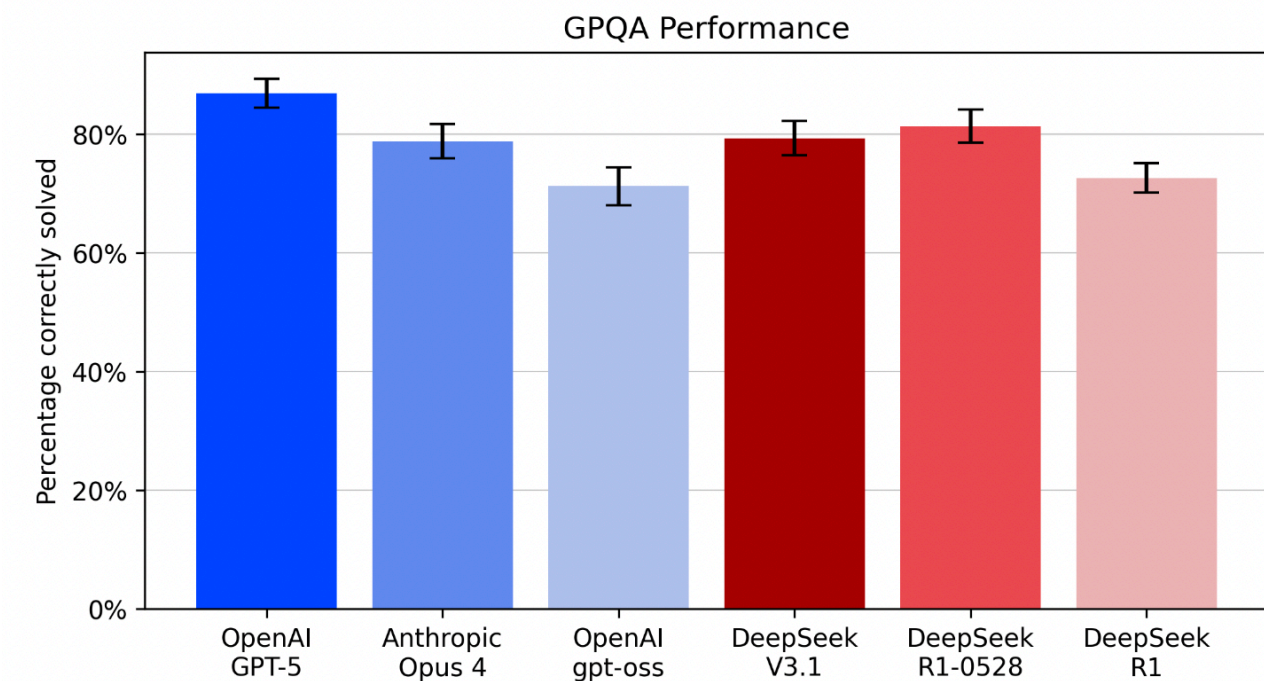
State what the eval measured, and where you're extrapolating.

Report uncertainty

A standard error alongside every metric — not a bare point estimate.

Share the protocol

Versioned settings, code, transcripts — enough to reproduce.



NIST's DeepSeek evaluation — a standard error on every score. Stage 3, worked.

The punchline: “None of that is a measurement problem.”

In NIST AI 800-2, where do most benchmark-evaluation failures happen?

- A Stage 1 — defining what you're actually measuring
- B Stage 2 — running the evaluation harness
- C Stage 3 — reporting the results
- D In the model weights themselves

Models A and B score 73% vs 68% raw. B has a higher guessing floor (0.35 vs 0.04) and beats A on hard items. What does that tell you?

- A** B is the stronger reasoner — it wins on the hard questions
- B** Most of B's hard-item correct answers are indistinguishable from hallucinations
- C** A and B are interchangeable for deployment
- D** B was simply tested on easier items

03

The Wrong Error Bar

Even with a valid construct, the standard-error formula most teams use is invalid the moment you run more than one trial per item. NIST AI 800-3 formalizes the fix.

Part 3 • benchmark-uncertainty

NIST Trustworthy and Responsible AI
NIST AI 800-3

Expanding the AI Evaluation Toolbox with
Statistical Models

Drew Keller
Kweku Kwegyir-Aggrey
Ryan Steed
Anita K. Rao
Julia L. Sharp
A. Stevie Bergman

This publication is available free of charge from:
<https://doi.org/10.6028/NIST.AI.800-3>

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STANDARDS AND TECHNOLOGY
U.S. DEPARTMENT OF COMMERCE

Two accuracies, two error bars

Benchmark accuracy

75.3%

NARROW CI

How well on *these* items? Uncertainty is LLM nondeterminism only — the right estimand when the question is whether a specific run passed.

Generalized accuracy

75.3%

WIDE CI

How well on the *superpopulation* the benchmark represents? Adds item-selection variance — what you actually care about when you deploy.

Same point estimate. Different claims. Different SEs. You don't deploy to answer 198 questions — you deploy to the domain they sample.

Why $\sqrt{n \cdot t}$ is the wrong denominator

$$SE_{\text{naive}} = \hat{\sigma} / \sqrt{n \cdot t}$$

With t trials on n items, the naive formula treats all $n \cdot t$ observations as independent. They're not — trials on the same item share its difficulty. Running a model on one question eight times is not eight different questions.

For **generalized accuracy** — the estimand that drives deployment decisions — the naive bars are **too narrow**. Comparisons that look significant at 5% may not be. The fix: use the per-item average as the unit of analysis (regression-free clustered SEs).

A Rasch model for benchmarks

$$Y_{i,j} \sim \text{Bernoulli}(\sigma(\theta_i - \delta_j)), \delta_j \sim N(0, \sigma^2)$$

θ_i — fixed effect

Capability of these specific, named models — what we want to estimate.

δ_j — random effect

Item difficulty, drawn from a superpopulation. Partial pooling borrows strength across items.

CIs come out **~0.013 narrower** without losing coverage. On GPQA-Diamond, AIC 33,995 vs. 44,378 — item difficulty is real and decisive. Classic IRT fixes items and samples people; here we fix models and sample items.

What the GLMM surfaces

Bad items

A BIG-Bench “Date Understanding” item scored $\delta = 7.58$ — every model wrong. On inspection it has two defensible answers. The GLMM flags it; an accuracy table hides it.

Domain structure

GPQA chemistry items run +0.59 vs. -0.55 for physics — across all 22 models. A structural property of the benchmark, invisible in one accuracy number.

Where to spend compute

ICC + effective samples/trial:
Movie-Rec has ICC 0.83, so 5 trials $\approx$ 1.15 effective observations.
Spend the budget on more items, not more trials.

What goes in the validation report

1 Declare the estimand

“75.3% generalized accuracy on GPQA-Diamond (n=198, t=8), 95% CI [73.1, 77.5], regression-free clustered SE.”
Not a bare “75.3%.”

2 Put the SE formula in the artifact

After 800-3, reporting $\sigma/\sqrt{(n \cdot t)}$ is a documented wrong choice, not an oversight.

3 Use θ for champion–challenger

Latent capability θ compares models across benchmarks of different difficulty — given explicit linking items.

“Which estimand, and which SE formula produced it?” now has a published answer.

04

The Judge and the Base Rate

An LLM judge looks reliable on a balanced test set, then quietly fails at the failure rates banking actually runs at — sub-1% prevalence, where a single judge cannot be a standalone detector and the dashboard cannot tell you whether failures are rare or invisible.

Two readings of one dashboard

A payments assistant handles **50,000 conversations a month**. An LLM judge reads every transcript for the failure the bank fears most — the agent waving through a scam payment. The dashboard reads **0.2% flagged**, stable for three months.

Reading 1 — the failure is rare

Scam-coaching genuinely almost never happens. The low, stable number is the system working as intended, and the queue is small because there is little to catch.

Reading 2 — the judge can't see it

The failures are there; the judge flags almost none of them. The same green dashboard, produced by an instrument that is blind to the thing it is supposed to catch.

The uncomfortable part: **the dashboard alone cannot tell you which.**

The base rate owns the flag, not the judge

$$\text{PPV} = \text{TPR} \cdot p / (\text{TPR} \cdot p + \text{FPR} \cdot (1-p)) = (0.90 \times 0.01) / (0.90 \times 0.01 + 0.05 \times 0.99) \approx 15\%$$

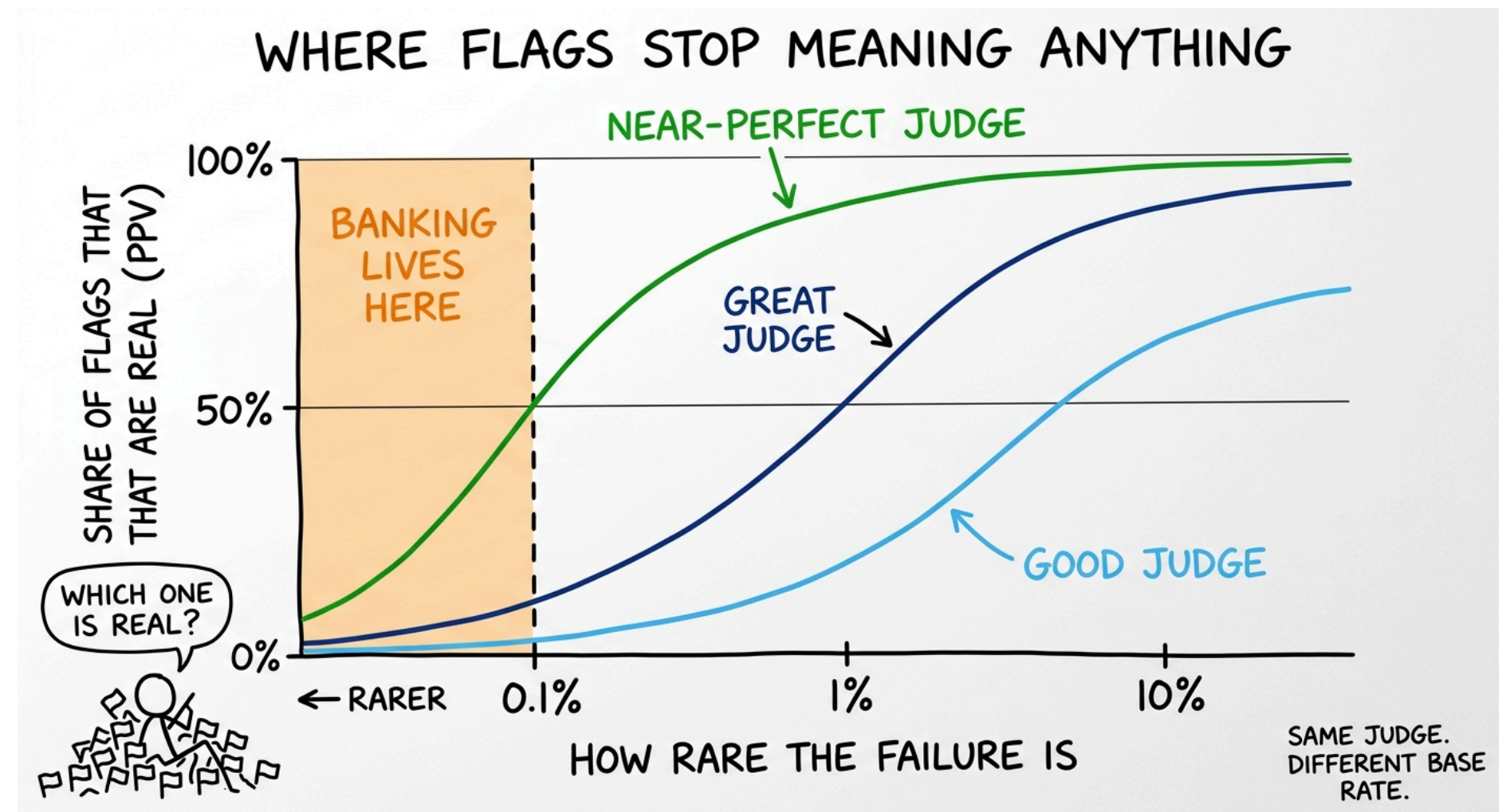
True failure rate	Good · 90/5	Great · 95/1	Near-perfect · 99/0.1
10%	67%	91%	99%
1% — where banking lives	15%	49%	91%
0.1%	1.8%	8.7%	50%

Columns are TPR / FPR. At 1% prevalence six of seven flags are false alarms — the false-positive paradox, owned by the base rate, not the judge. Even a near-perfect judge flips a coin at 0.1%, and no prompt tuning moves it.

Where flags stop meaning anything

The same judge produces a usable review queue at 10% prevalence and pure noise at 0.1%. Nothing about the judge changed — only the **base rate** did. Banking lives on the rare end of this curve, where every judge quality collapses toward zero.

Share of flags that are real (PPV) vs. how rare the failure is



Judges fail the direction you don't guess

The table assumed false alarms. The measured evidence says deployed judges fail the other way — **missing the rare positives**. Agreeableness bias: judges reliably confirm and unreliably reject.

Real Mistake

GPT-4's recall on its own errors fell to 7–12% against a human F1 of 95.7. Self-consistency did nothing; majority voting across 12 runs often made it worse.

14 frontier judges

Judging code feedback, judges caught under 25% of true failures; 26% were missed by every one of the 14. TPR correlates negatively with capability ($\rho \approx -0.77$).

Production agent

A deployed ordering agent's judge caught ~22% of confirmed defects — zero of 100 rounds in one batch — undercounting the true defect rate 3–6×.

And a blind spot no confusion matrix measures: the judge grades only what entered the transcript — never the evidence the agent didn't retrieve or the check that never ran.
“Complete with respect to what it sees.”

Measurement and detection aren't the same job

"What is our failure rate?" — solvable

Treat the judge as a noisy instrument: estimate its TPR/FPR on a gold set and propagate the error. Rogan-Gladen recovers an unbiased prevalence from biased counts; prediction-powered inference tightens the interval $\sim 3\times$. Naive raw-flag-rate intervals see coverage collapse to zero under judge bias.

"Catch this failure" — not solvable that way

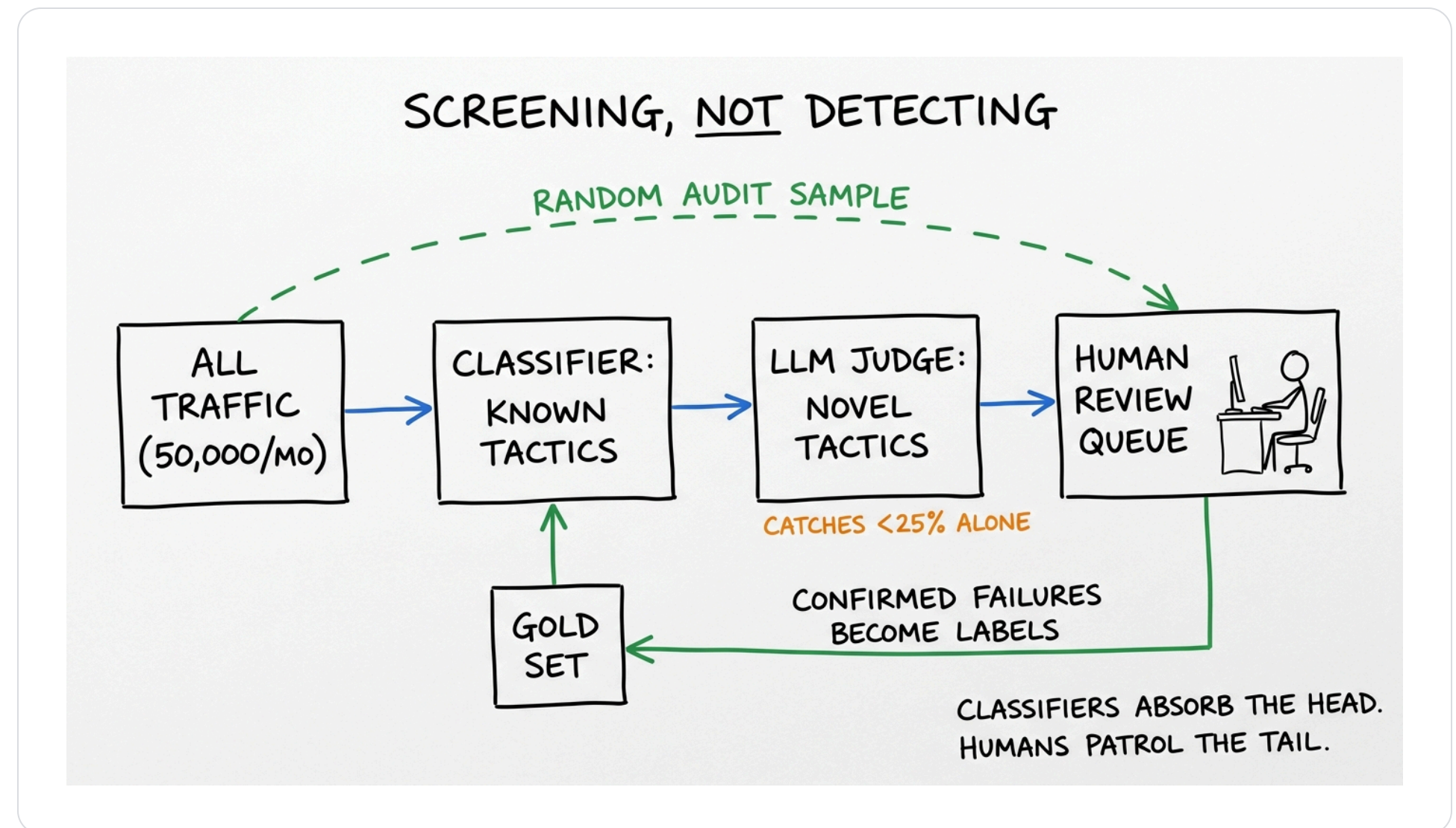
Correction adjusts the aggregate; it cannot reach into 50,000 transcripts and tell you which one was the scam. A corrected rate is a monitoring metric. Stopping the payment before it settles is a control — different math, different artifact.

Most monitoring decks conflate the two: a judge-produced "failure rate" shown as both evidence the rate is low *and* the mechanism that catches failures. At 1% it is defensibly neither — until the gold-set work is done.

Give each layer a job it can actually do

No single judge works at this base rate; a cascade does. A **classifier** screens known tactics, the **LLM judge** patrols for novel ones at high TPR, and humans confirm. Confirmed failures feed a gold set; a **random audit stream** is the only input that surfaces what the whole pipeline misses.

Classifiers absorb the head · humans patrol the tail



THE WORKED EXAMPLE · 0.2% OF 50,000 = 100 REAL FAILURES

The review queue is a checkpoint someone designs

Judge	Flags/mo	PPV	Reviews/day	Caught / 100	Missed
The one you think you deployed · 90 / 5	≈2,585	3.5%	≈120	90	10
The one as measured · 25 / 4	≈2,021	1.2%	≈95	25	75

The measured judge is worse on both axes: a ~95-review-a-day queue that still misses three of four failures. A reviewer clearing dozens of flags to find one real case is in the alert-fatigue regime — and the FSB's June 2026 consultation now tells supervisors to investigate humans who **approve agent recommendations without modification**.

Zero flags isn't zero failures. At these base rates, it isn't even evidence of it.

The work the field skips

- 01 Metrics aren't measurement** Psychometrics is the missing discipline — construct validity, not dashboards.
- 02 NIST 800-2 gives the process** “Emerging practice” is a map of where the field still fails.
- 03 NIST 800-3 gives the statistics** The leaderboards almost certainly report the wrong error bar.
- 04 A judge isn't a detector** At banking prevalence: screen with a pipeline, correct the rate, never trust a single flag.

It's institutionally uncomfortable — not technically hard.